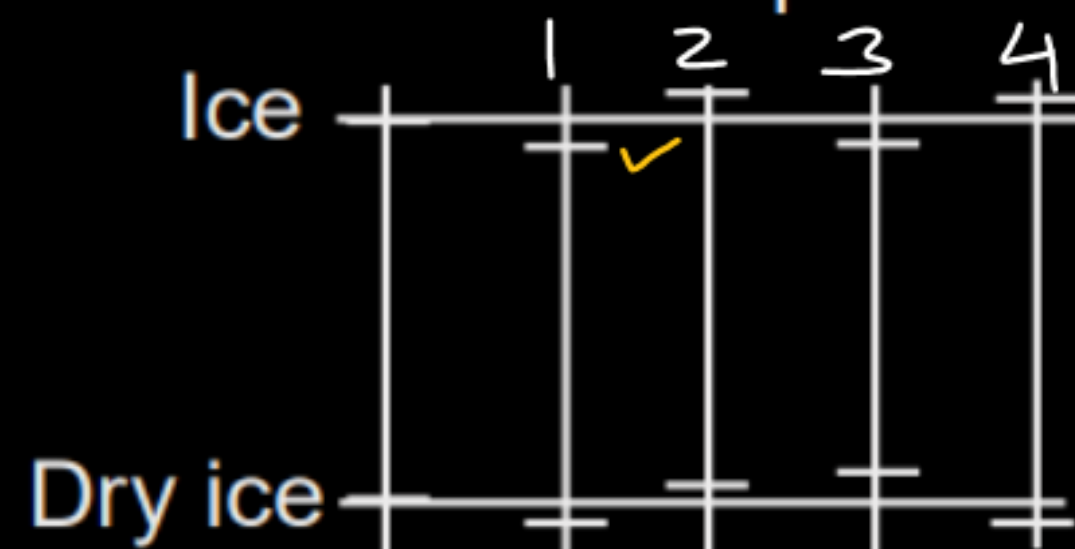


The first vertical line in the diagram represents a thermometer with the melting points for ice and dry ice. (melting point of ice is greater than that of dry ice) at normal atmospheric pressure. The numbered vertical lines represent possible melting points of both the substances at 10 atmospheric pressure. Which vertical line best represents the melting points of both the substances at 10 atmospheric pressure relative to the respective values at normal atmospheric pressure?

Note : The difference in temperatures are not to the scale.



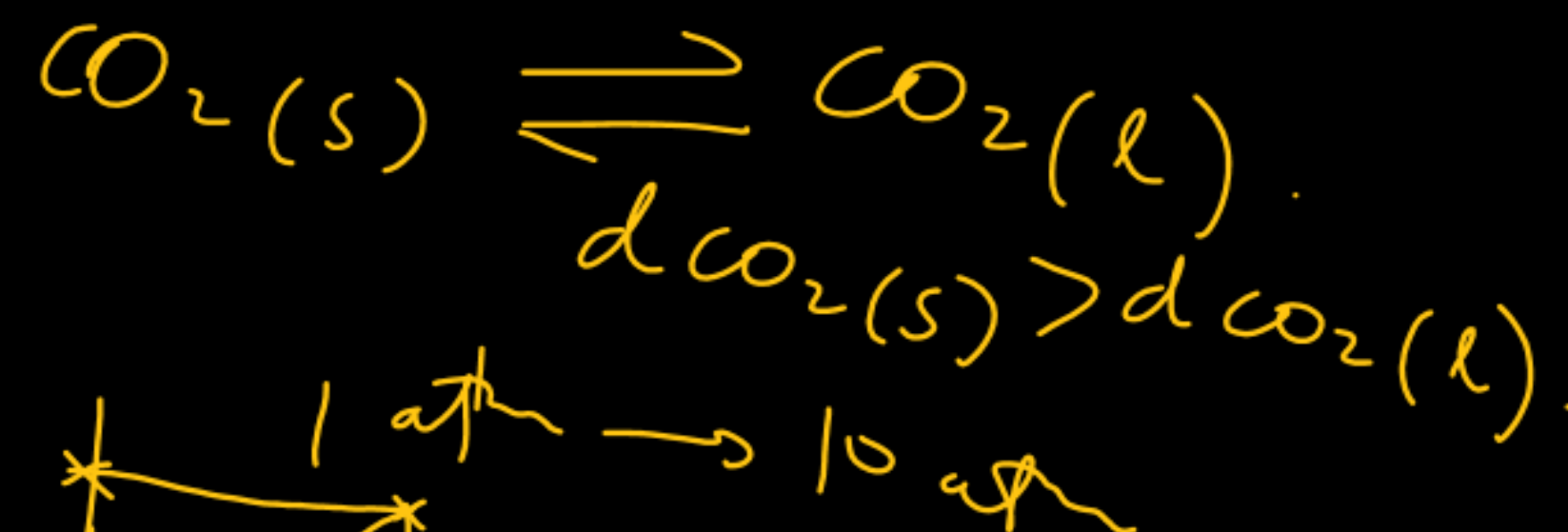
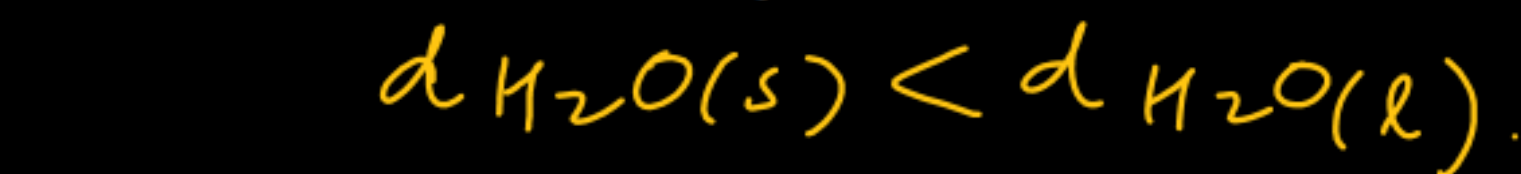
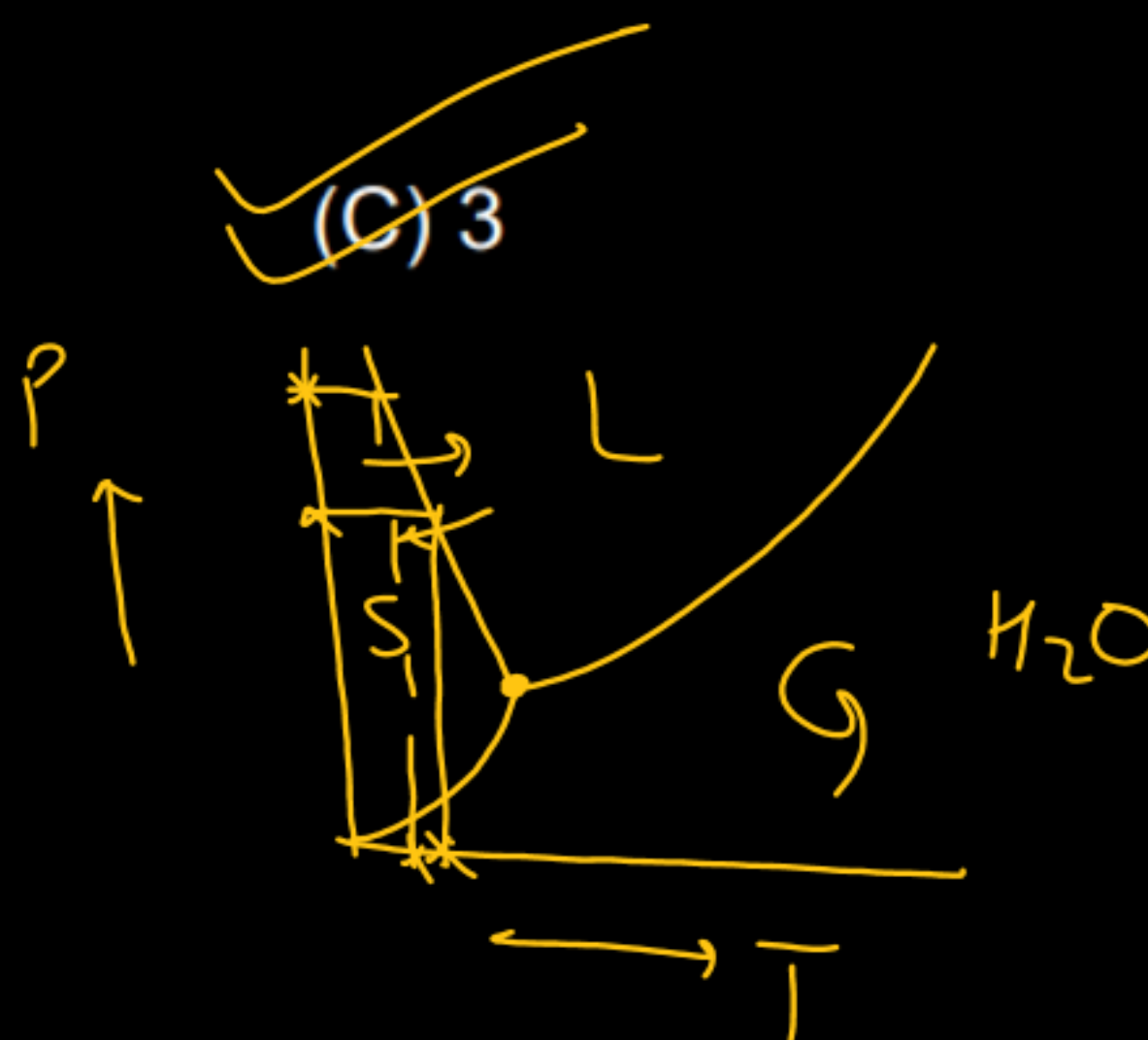
(A) 1

(B) 2

(C) 3

(D) 4

ice, m.p ↓
dry ice, m.p ↑



Which of the following statements is correct for gas A having molar mass 16g and density 0.75 g/L at 2 atm pressure and 27°C temperature.

- (A) Force of attraction are dominating than force of repulsion among the gas molecules.
- ✓ (B) Force of repulsion are dominating than force of attraction among the gas molecules.
- (C) Gas molecules are behaving ideally
- (D) Nothing can be said with certainty

Real gas, $d_{\text{real gas}} = 0.75 \text{ g/L}$.

$$PM = Z \times dRT$$

$$Z = \frac{2 \times 16}{0.75 \times \frac{1}{12} \times 300}$$

$$= \frac{2 \times 16}{\left(\frac{3}{4}\right) \times 25} = \frac{32 \times 4}{3 \times 25} = \frac{128}{75} > 1$$

$Z > 1$, indicates repulsive forces dominant.

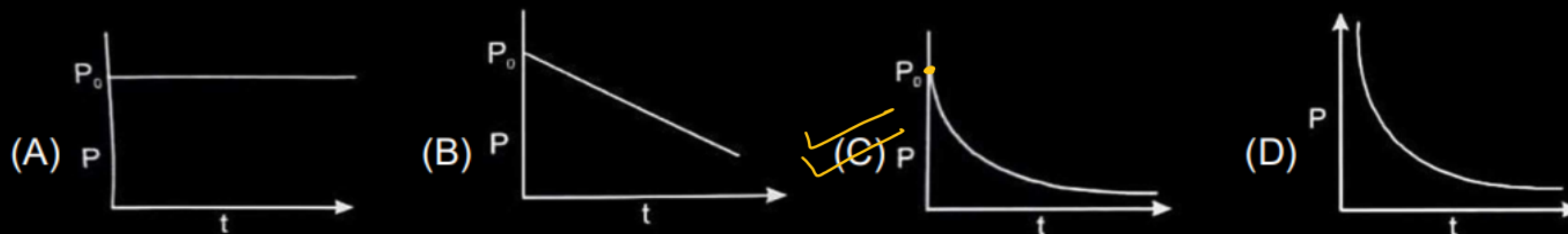
Graham's law tells us about rate of effusion or diffusion of gases. In modern form, it simply states that the rate of effusion or diffusion of any ideal gas is inversely proportional to square root of its molar mass.

$$\frac{r_1}{r_2} = \frac{P_1}{P_2} \sqrt{\frac{M_2}{M_1}}$$

$$\Rightarrow r \propto \frac{P}{\sqrt{M}} \Rightarrow -\frac{dp}{dt} = k \times \frac{P}{\sqrt{M}} \Rightarrow \int_{P_0}^P \frac{dp}{P} = \frac{k}{\sqrt{M}} \int_0^t dt$$

$$\Rightarrow -\ln \frac{P}{P_0} = \frac{k t}{\sqrt{M}} \Rightarrow \ln \frac{P_0}{P} = \frac{k t}{\sqrt{M}}$$

→ 10 moles of N_2 is placed in a vessel of constant volume and temperature. A hole is punctured in the vessel and left in vacuum. The pressure of N_2 in the vessel will vary with time as

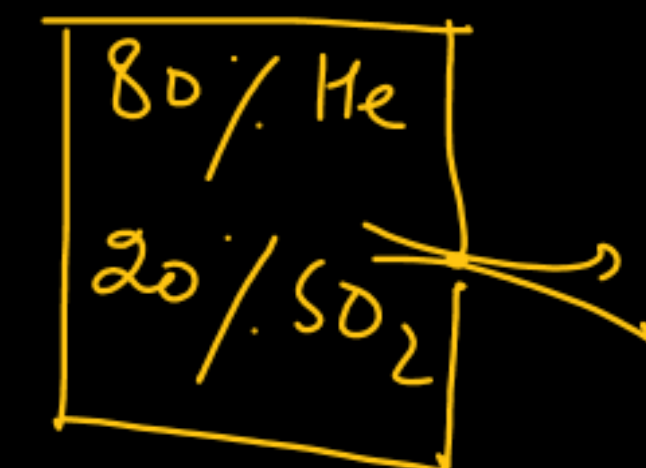


$$P = P_0 e^{-\frac{k t}{\sqrt{M}}}$$

initial P_0

→ If a mixture of (80 mol %) He and (20 mol %) SO_2 is taken in an isothermal isochoric vessel, as the gases leak through a small hole in the vessel. When placed in vacuum $T, V \rightarrow$ constant

- (A) Average molar mass of gas mixture increases with time
- (B) Average molar mass of gas mixture remains same with time
- (C) Average molar mass of gas mixture decreases with time
- (D) Variation of M_{avg} with time depends on temperature as well as area of hole, so it cannot be commented upon in general.



Two moles of CO_2 gas (treated to be an ideal gas) expands isothermally (in thermal contact with surroundings, temperature = 27°) against a constant external pressure of 1 bar. The initial and final volumes of the gas are 10 litre and 30 litre, respectively. [$R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$, $\ln 10 = 2.3$, $\log 2 = 0.3$, $\log 3 = 0.48$]

Imp

Select correct option regarding change in entropy of system (ΔS_{sys}) and of the surrounding (ΔS_{surr}):

(A) $\Delta S_{\text{sys}} > 0$, $\Delta S_{\text{surr}} = 0$

(C) $\Delta S_{\text{sys}} = 0$, $\Delta S_{\text{surr}} = 0$

~~(B)~~ $\Delta S_{\text{sys}} < 0$, $\Delta S_{\text{surr}} > 0$

~~(D)~~ $\Delta S_{\text{sys}} > 0$, $\Delta S_{\text{surr}} < 0$

$$\Delta S_{\text{sys}} = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$$

$$T_1 = T_2$$

$$\Delta S_{\text{sys}} = nR \ln \frac{V_2}{V_1}$$

$$= 2R \times \ln \frac{30}{10}$$

$$= +ve$$

The change in entropy of universe in above isothermal process is :

(A) 20.54 JK^{-1}

~~(B)~~ 11.67 JK^{-1}

(C) 18.33 JK^{-1}

(D) 15.5 JK^{-1}

$$\Delta S_{\text{surr}} = \frac{q_{\text{surr}}}{T}$$

$$= -\frac{q_{\text{sys}}}{T}$$

$$= -\frac{w}{T}$$

$$= -\frac{-ve}{T}$$

$$= +ve$$

$$\Delta U = q + w$$

$$= 0$$

$$U = f(T)$$

$$\text{for an ideal gas.}$$

Irr. iso. process.

$$\Delta S_{sys} = nR \ln \frac{V_2}{V_1}$$

$$= 2 \times 8.3 \times \ln \frac{30}{10}$$

$$= 2 \times 8.3 \times 2.3 \times \log 3$$

$$= 8.3 \times 4.6 \times 0.48$$

$$\cong 8.3 \times 2.3$$

$$\cong 19.1 \text{ J/K}$$

$$\Delta S_{surr} = \frac{Q_{surr}}{T_{surr}} = - \frac{Q_{sys}}{T_{surr}}$$

Irr. iso. process, $\Delta U = Q + W$

$$Q = -W_{irr}$$

$$Q_{irr} = +2500 \text{ J}$$

$$\Delta S_{surr} = - \frac{2500}{300} = -6.67 \text{ J/K}$$

$$\begin{array}{r} 8.3 \\ 2.3 \\ \hline 249 \\ 166 \\ \hline 19.09 \end{array}$$

$$\begin{array}{r} 19.1 \\ 6.7 \\ \hline 12.4 \end{array}$$

$$\begin{array}{l} \Delta S_T = 19.1 - 6.7 \\ \hline \hline \hline \cong 12.4 \text{ J/K} \end{array}$$

$$(W_{rev})_{iso} = -nRT \ln \frac{V_2}{V_1}$$

$$(W_{irr})_{iso} = - \int_1^2 P_{ext} \times dV$$

$$= -P_{ext} \times \Delta V$$

$$= -1 \times 10^5 \text{ Pa} \times (30 - 10) \times 10^{-3}$$

$$= -20 \times 100 \text{ Joule}$$

$$= -2000 \text{ Joule}$$

Ajay feels his intelligence increases on eating desi ghee. Moreover, the empty 'tin' of Ghee can be used for storing homemade sweets also!!

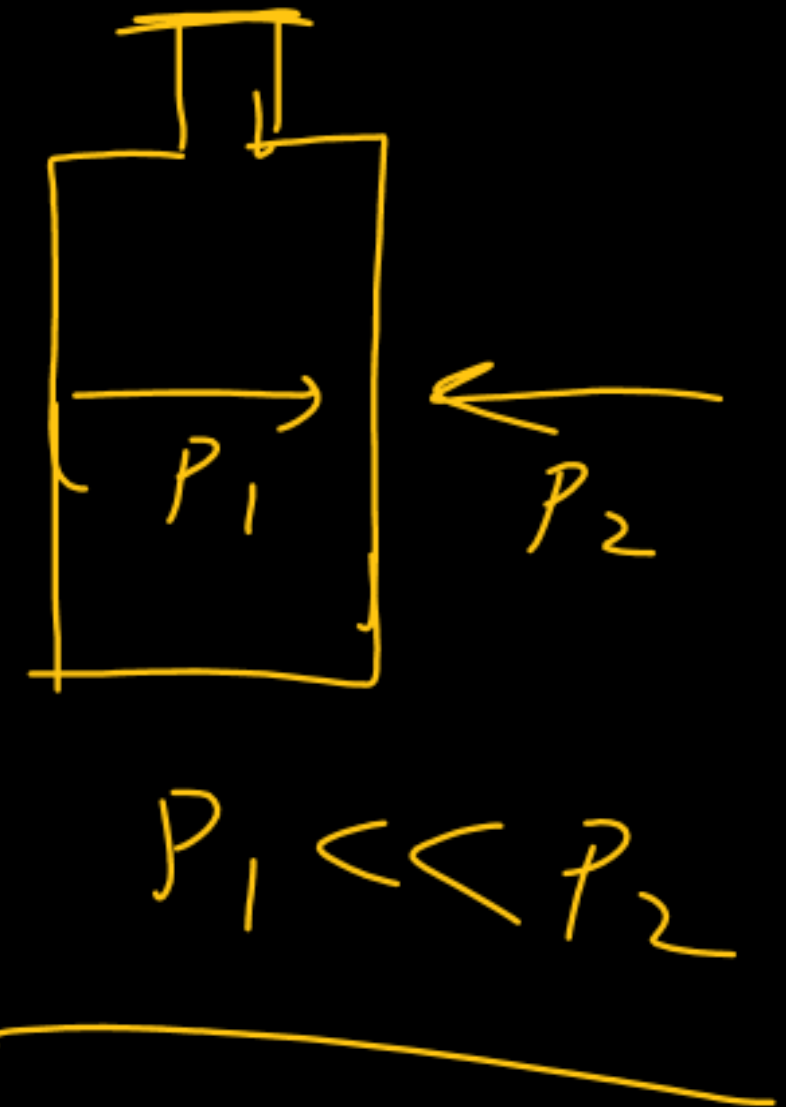
Ajay tries to clean 15 L 'tin' of desi Ghee (made up of tin metal) with boiling water by the following possible procedures:

- ✓(x) Some amount of boiling water is added to it, shaken well and its cap is tightly replaced. It is then thrown in cold water.
- ✓(y) Water is filled in it partially and kept on stove. As the water begins to boil, it is allowed to continue for sometime. The cap is then tightly replaced and the tin is thrown in cold water.
- (z) Water is filled in it partially. Its cap is tightly replaced. It is then kept on stove till water gets heated to 100°C . After sometime, the vessel is shaken vigorously and then thrown in cold water.

Unfortunately, the tin gets crushed/damaged in one or more of the above cases. Assume that it does not explode in any case.

In which procedure(s) the tin will not get crushed?

- (A) x and y (B) x only ✓(C) z only (D) y and z



Ans: 6

$$p_{N_2} = 0.5p, p_{O_2} = 0.3p, p_{He} = 0.2p$$

Scuba divers use a gas mixture containing 50% N_2 , 30% O_2 and 20% He for breathing. If this gas is used in breathing apparatus, the mole ratio of dissolved N_2 , O_2 and He in the blood is $p : q : r$ (minimum integer ratio). Find $(p + q + r)/10$, rounded to closest integer. Given k_H values for $O_2 = 150$ bar, $N_2 = 90$ bar, $O_2 = 50$ bar at body temperature for solubility in blood.

$p \rightarrow \text{Total } p_r$

$$X = \frac{n}{n_T}$$

$$X_{N_2} : X_{O_2} : X_{He} = \frac{0.5p}{90} : \frac{0.3p}{50} : \frac{0.2p}{150}$$

$$n_{N_2} : n_{O_2} : n_{He} = \frac{0.5}{9} : \frac{0.3}{5} : \frac{0.2}{15}$$

$$= \frac{5}{9} : \frac{3}{5} : \frac{2}{15}$$

$$= 5 \times \frac{15}{5} : 3 \times \frac{27}{3} : 2 \times \frac{9}{3}$$

$$= 25 : 27 : 6$$

Henry's law,

$$p_A = X_A \times (K_H)_A$$

$$X_A = \frac{p_A}{K_H}$$

$$\frac{p+q+r}{10} = \frac{25+27+6}{10} = \frac{58}{10} = 5.8 \approx 6$$

Ans: 8

Dissolving 1 mole of camphor ($C_{10}H_{16}O$) in one mole of an equimolar mixture of acetophenone(A) ($P_A^0 = 600$ torr) and benzophenone(B) ($P_B^0 = 400$ torr) fails to change its vapour pressure. However, adding x moles of (non-volatile) anthracene to this solution lowers its vapour pressure to 200 torr. If vapour pressure of pure camphor is $100y$ (torr) find $(x+y)$.

Solⁿ: Camphor $\rightarrow$ 1 mol. (v.p of Camphor = 500 torr)
 { A $\rightarrow$ 0.5 mol. (600 torr)
 { B $\rightarrow$ 0.5 mol. (400 torr)

$$500 = 100y$$

$$y = 5$$

$\rightarrow$ v.p. of mixture of A & B = $0.5 \times 600 + 0.5 \times 400$
 $= 500$ torr.

x mol $\rightarrow$ anthracene, v.p = 0

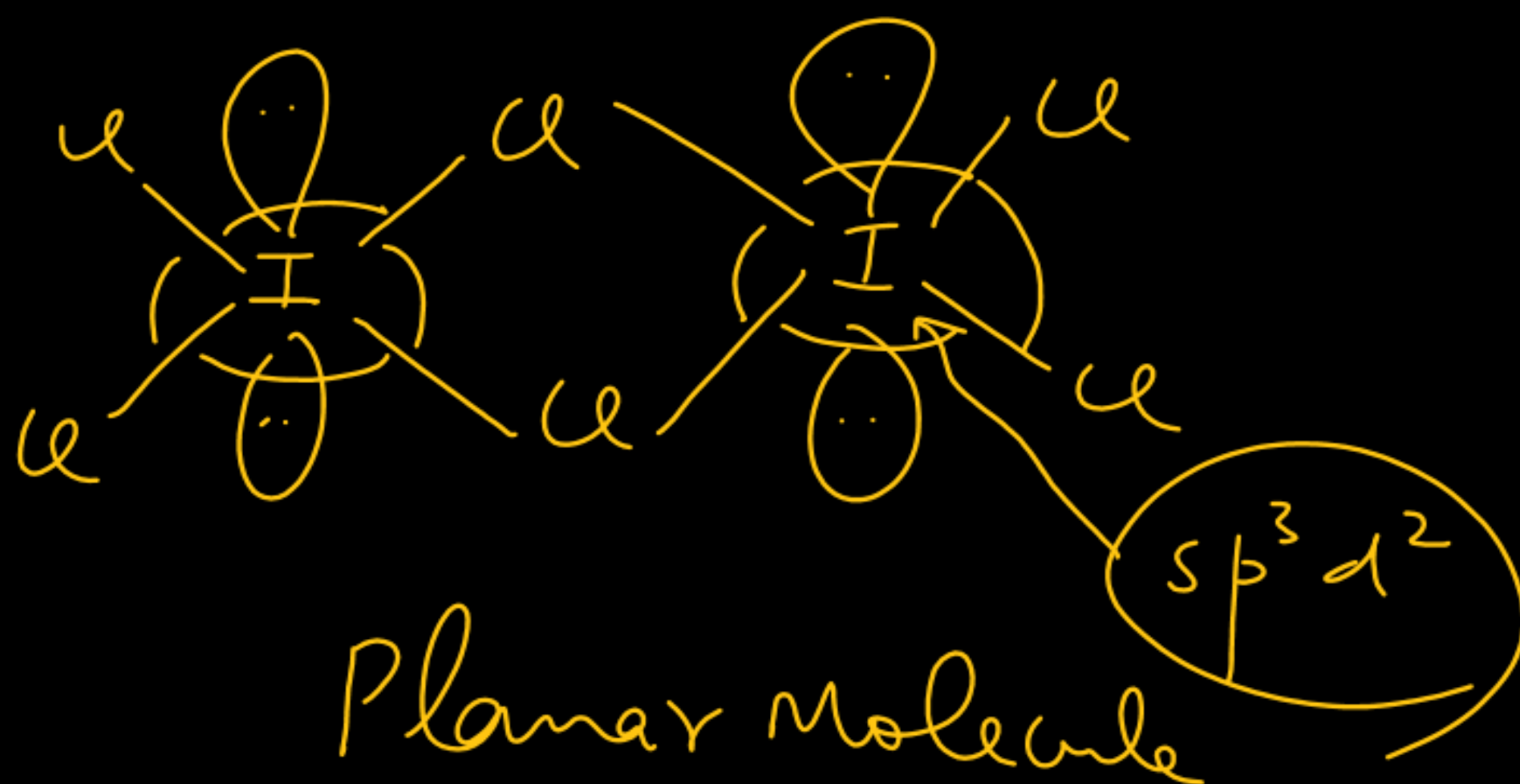
$$p = x_A p_A^0 + x_B p_B^0 + \dots$$

$$\begin{aligned} \text{Total v.p} = 200 &= \frac{1}{1+0.5+0.5+x} \times 500 + \frac{0.5}{1+0.5+0.5+x} \times 600 + \frac{0.5}{1+0.5+0.5+x} \times 400 \\ (x+y) &= 3 + 5 \\ &= \underline{8} \text{ ans.} \end{aligned}$$
$$= \frac{1}{2+x} (500 + 300 + 200) = \frac{1000}{2+x} \Rightarrow 2+x = 5$$
$$\underline{x = 3}$$

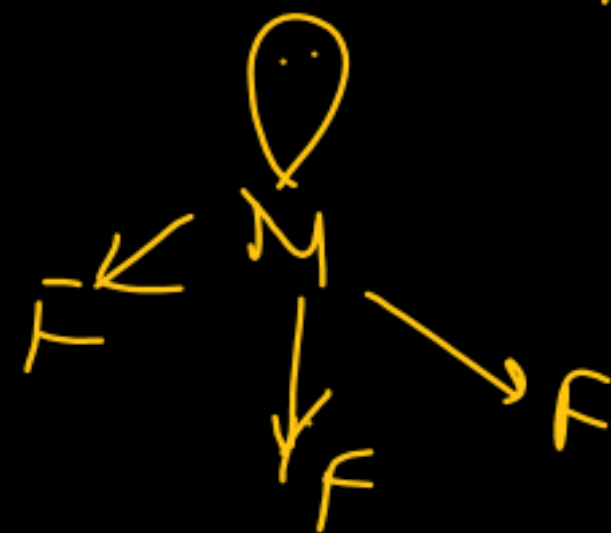
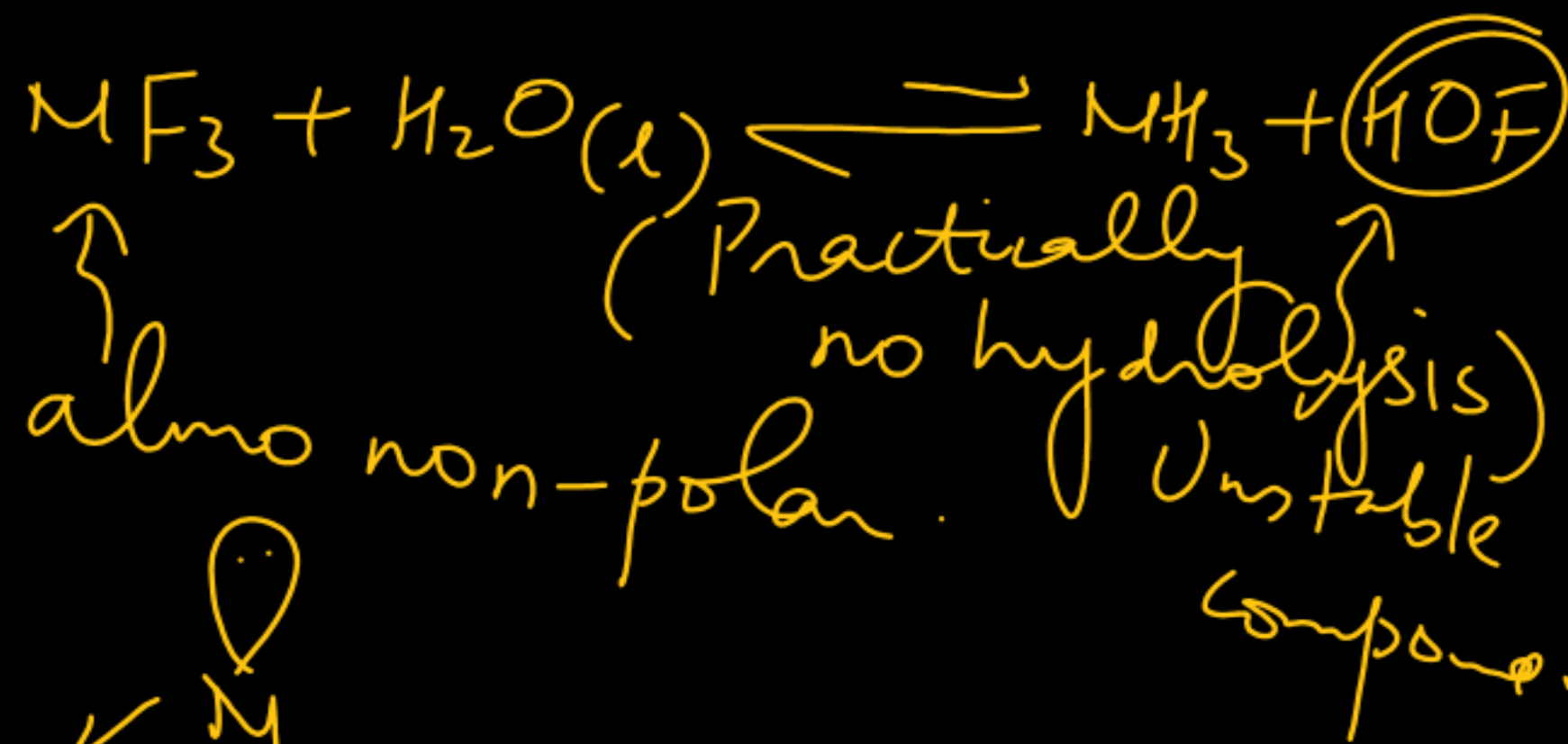
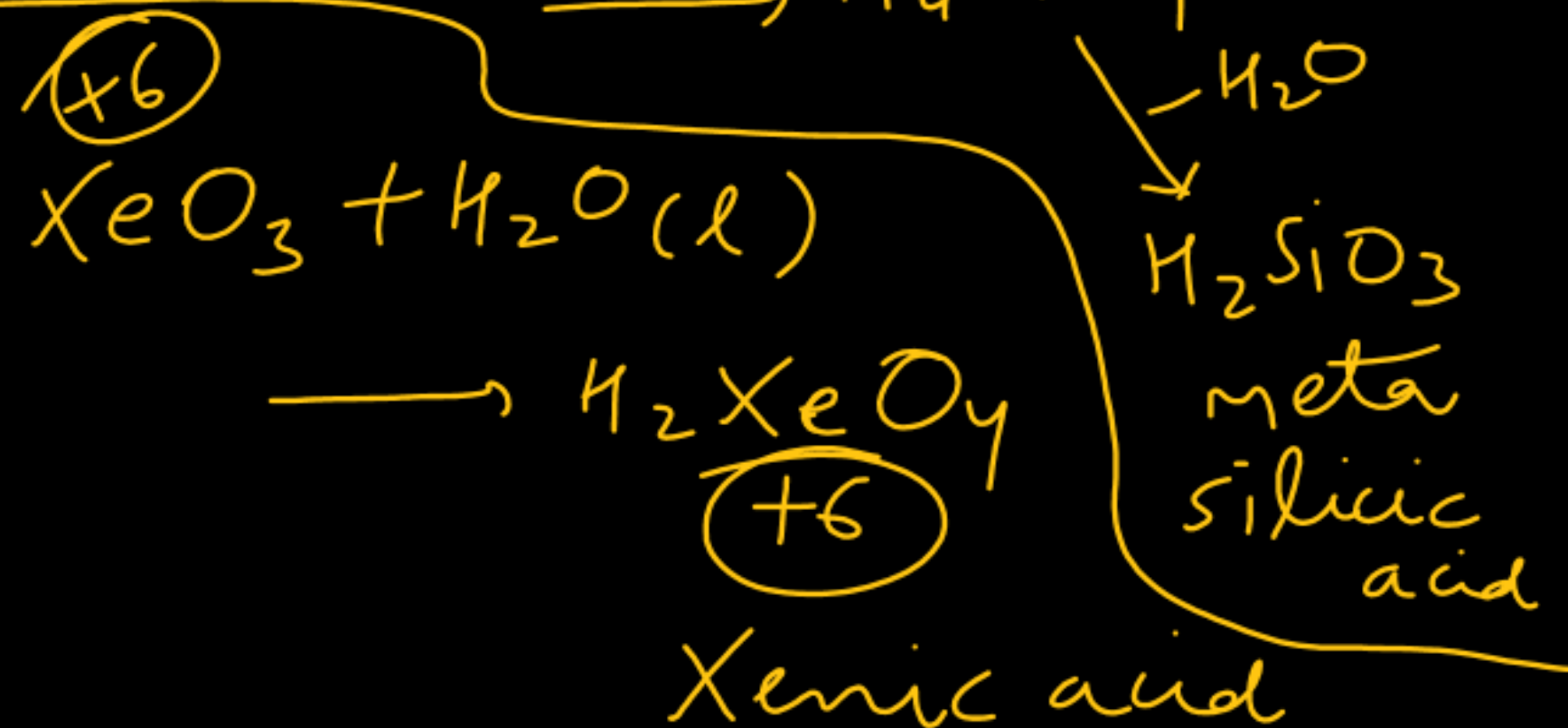
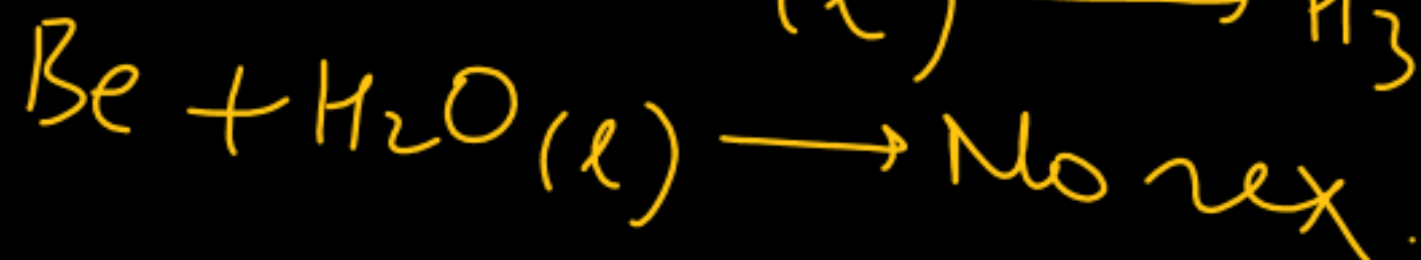
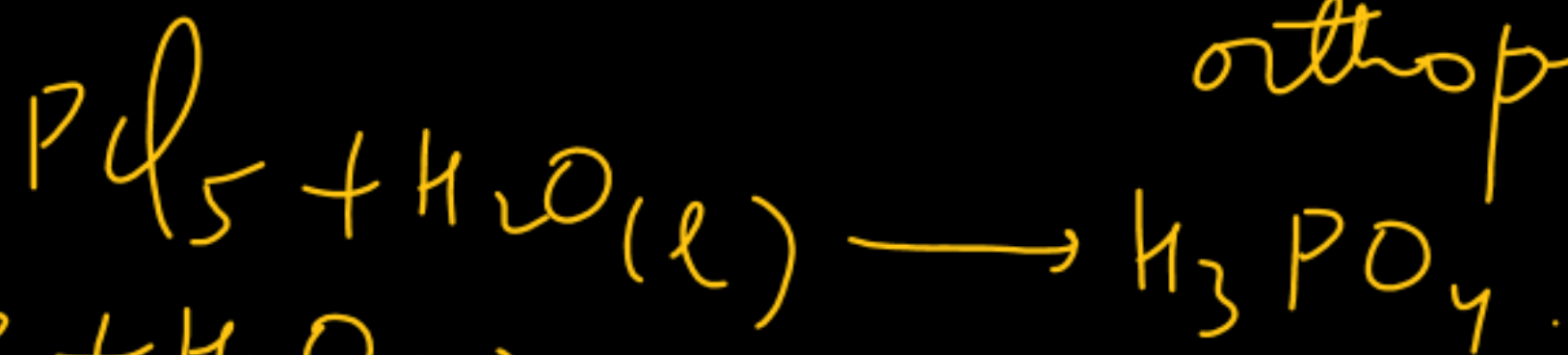
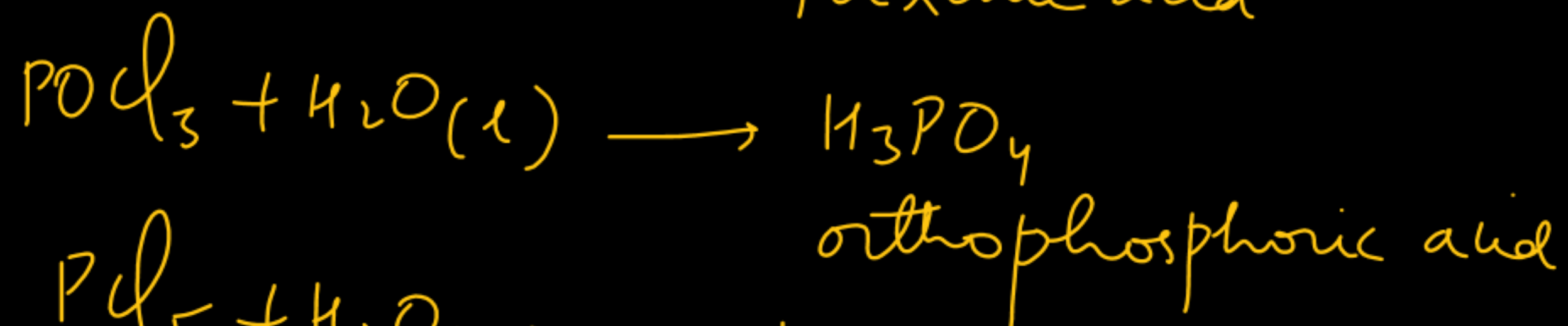
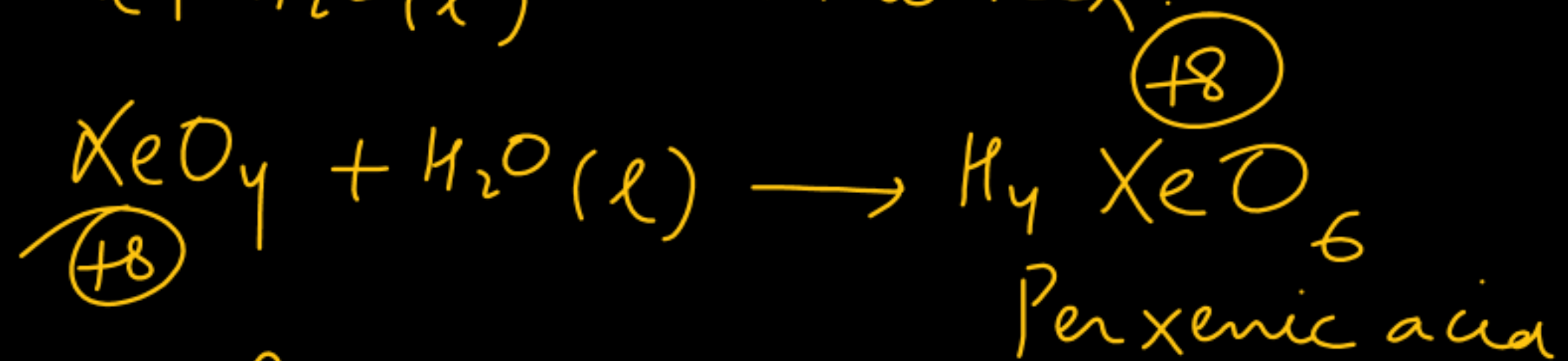
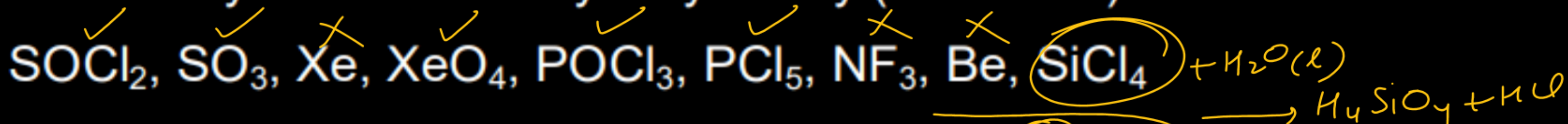
ICl_3 is an orange yellow colored solid that dimerizes in solid state as I_2Cl_6 . Based on VSEPR theory, number of 90 degree $\text{Cl} - \text{I} - \text{Cl}$ bond angles is in the dimeric species. Neglect any minor deviations from ideal bond angle.

$\text{I}_2\text{Cl}_6 \longrightarrow$ dimer of ICl_3 .

Ans : 8



How many of these are hydrolysed by (react with) water?



How many of the following will turn red litmus blue?

base

$\text{NH}_4\text{Br}(\text{aq})$, ~~acidic~~ X

$\text{Sr}(\text{NO}_3)_2(\text{aq})$, neutral ✓

$\text{HCOONa}(\text{aq})$ ✓



$\text{CH}_3\text{COONH}_4(\text{aq})$, X

$\text{ZnCl}_2(\text{aq})$, acidic X

$\text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}(\text{aq})$ X



$\text{NH}_3(\text{aq})$, ✓

$\text{KI}(\text{aq})$, X

$\text{CH}_3\text{COOH}(\text{aq})$ X

acidic

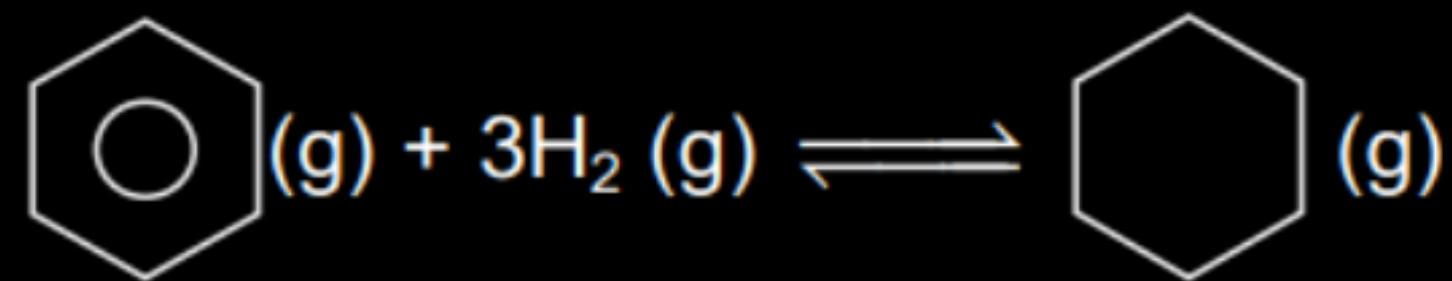
basic

almost neutral

Ans: 2

A rigid platinum vessel of volume 24.63 L contains H_2 gas at 3 atm pressure at 300 K. As 1 mole of benzene (liquid) is added to it, pressure rises to 3.2 atm at 300K. At this temperature, no chemical reaction of any type occurs.

Now, temperature is raised to 360 K. All the liquid vaporizes entirely. Benzene is also catalytically hydrogenated to cyclohexane partially:



$$PV = nRT$$

$$3 \times 24.63 = n_{\text{H}_2} \times 24.63 \Rightarrow n_{\text{H}_2} = 3 \text{ mol}$$

$$(V \cdot P)_{\text{C}_6\text{H}_6, 300\text{K}} = 3.2 - 3 = 0.2 \text{ atm}$$

Pressure in the system at equilibrium is measured to be 2.4 atm, at 360 K. It is then cooled back to 300 K. The pressure in the vessel falls to 1.25 atm. Some amount of liquid also appears. Given that vapour pressure of cyclohexane is 0.3 atm at 300 K, the mole fraction of benzene in this liquid is x_B .

x_B is: 0.5

at 360K

$$\text{C}_6\text{H}_6(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons \text{C}_6\text{H}_{12}(\text{g})$$

1 mol	3 mol	0
$1-x$	$3-3x$	x mol
$\frac{1}{3}$ mol	1 mol	$\frac{2}{3}$ mol

at eq^m

→ when it is cooled to 300K, C_6H_6 & C_6H_{12} are liq. now.

$PV = nRT$

$$\frac{2.4}{2} \times 24.63 = (1-x + 3-3x+x) \times 0.082 \times 300$$

$$2 = 4 - 3x$$

$$3x = 2$$

$$x = \frac{2}{3} \text{ mol}$$

~~$$(P_T)_{300K} = 1.25 = (V.P)_{C_6H_6} + (V.P)_{C_6H_{12}} + p_{H_2}$$~~

~~$$(p_{H_2})_{300K} \times 24.63 = 1 \times 24.63$$~~

$$(p_{H_2})_{300K} = 1 \text{ atm}$$

$$p_T = 1.25 = (V.P)_{\text{solution}} + p_{H_2} \rightarrow 1 \text{ atm}$$

$$(V.P)_{\text{solution}} = 1.25 - 1 = 0.25 = x_{\text{benz}} \times p^{\circ}_{\text{benz}} + x_{C_6H_{12}} \times p^{\circ}_{C_6H_{12}}$$

$$0.25 = x_b \times 0.2 + (1 - x_b) \times 0.3$$

$$0.25 = 0.2x_b + 0.3 - 0.3x_b$$

$$\Rightarrow \boxed{x_{\text{Benz}} = 0.5}$$

$$\therefore 0.1x_B = 0.3 - 0.25 = 0.05$$